

NutriScore AI (India) — Technical & Operational Whitepaper

Comprehensive Staff Training & Implementation Guide: System Architecture, Algorithm Engines, Feature Breakdown, Mathematical Formulations, and Clinical Nutrition Rules.

1. Executive Summary & Why NutriScore-India Exists

Traditional nutrient profiling models (such as the French/European FSA Nutri-Score) were engineered for Western dietary patterns. When applied to traditional Indian cooking, European algorithms fail catastrophically: they heavily penalize healthy Indian staples (e.g. cold-pressed oils, lentils) while ignoring India's critical public health burdens: **protein quality bottlenecks (DIAAS)**, **micro-nutrient deficiencies (anemia)**, and **diabetes/hypertension risks**.

NutriScore-India is the country's first algorithmic nutritional profiling system developed **exclusively on official Indian government benchmarks**: the *Dietary Guidelines for Indians (ICMR-NIN 2024)*, the *Revised 2024 Recommended Dietary Allowances (RDA)*, and the *Indian Food Composition Tables (IFCT 2017)*. Foreign models have been completely eliminated.

Key Takeaway for Staff: NutriScore-India does not merely assign an arbitrary letter grade. It acts as an **Intelligent Clinical Dietitian & Culinary Advisor**, evaluating biological protein quality, screening for chronic disease risks, and offering actionable, kitchen-ready recipe adjustments.

2. The Two Scientific Profiling Modes Explained

The system provides two distinct profiling operational modes selectable right from the top header:

Evaluation Dimension	■ PPQND 7-Tier (Recommended)	■ Full IFCT 39-Nutrient Standard
Full Name	Personalised Protein Quality & Nutrient Density	Indian Food Composition Tables 39-Nutrient Model
Primary Purpose	Individual & family meal planning, clinical dietetics, recipe optimization.	Food laboratories, academic research, industrial packaged food formulation.
Demographic Calibration	11 Dynamic Life Stages (Adult Male, Adult Female, Pregnant, Lactating, Adolescents, Children, Elderly, Athletes).	National Adult Standard (Adult Moderate Work Baseline, 65kg reference man).
Protein Analysis	Full DIAAS & 9 Essential Amino Acids + Limiting Amino Acid bottleneck detection + Cereal-Pulse synergy index.	Crude Total Protein Quantity (g) assessed against daily recommended reference allowance.
Scoring System	7-Tier Grade (\$A+, A, B, C, D, E, F\$) based on Benefit-Risk mathematical balance equation.	5-Tier Grade (A, B, C, D, E) standard 5-color Nutri-Score.
Output Panels	4 Interactive Diagnostic Panels + Tailored Kitchen Balance Tips & Score Improvement Forecasts.	Comprehensive Nutrient Contributors Breakdown (Top 5 positive & negative drivers).

3. Feature-by-Feature Operational Guide (How Each Works)

Feature 3.1: 11 Demographic Life Stages Selector (ICMR-NIN 2024)

A recipe cannot have the same health impact for an adult male, a pregnant mother, or a 2-year-old child. The system calibrates all nutrient requirements dynamically according to the chosen demographic profile:

Demographic Profile	Reference	Protein RDA	Iron RDA	Key Clinical Priority (ICMR-NIN 2024)
---------------------	-----------	-------------	----------	---------------------------------------

Adult Male	19–59y (65kg)	54.0 g	19.0 mg	Metabolic baseline, maintenance of lean skeletal mass.
Adult Female	19–59y (55kg)	46.0 g	29.0 mg	Elevated Iron RDA to combat reproductive-age anemia.
Pregnant Mother	2nd / 3rd Tri	67.0 g	27.0 mg	High anabolic demand, fetal tissue growth, folate & zinc.
Lactating Mother	0–6 months	63.0 g	23.0 mg	Breastmilk synthesis, elevated calcium (1200mg) and vitamin A.
Adolescent Female	14–18y	46.0 g	32.0 mg	Highest national iron requirement during pubertal growth spurt.
Adolescent Male	14–18y	55.0 g	22.0 mg	Rapid lean muscle accretion, peak bone mineral density.
Senior Citizen	> 60 years	52.0 g	19.0 mg	Sarcopenia prevention, strict sodium & saturated fat moderation.
Active Athlete	High Demand	85.0 g	25.0 mg	Elevated BCAAs (Leucine, Valine) for muscle protein synthesis.

Feature 3.2: Protein Quality & Amino Acid Matrix (40% Total Weight)

In Indian diets, assessing only crude protein grams is clinically inadequate because plant proteins have limiting amino acids. According to **Liebig's Law of the Minimum**, protein synthesis in the human body is capped by whichever essential amino acid is present in the lowest concentration.

1. **Complementary Protein Matrix:** Evaluates traditional Indian culinary pairings that overcome single-source amino acid deficits:
- *Cereal + Pulse (Rice + Dal, Roti + Chana) [Synergy: 8/10]:* Lysine in pulses balances Lysine deficiency in cereals.
 - *Pulse + Dairy (Palak Paneer, Dal + Curd) [Synergy: 9/10]:* Dairy adds complete bioavailable sulphur amino acids.
 - *Single Cereal Grain alone (Plain Rice / Plain Bread) [Synergy: 2/10]:* Penalized due to severe Lysine bottleneck.
2. **Smart Auto-Estimation Checkbox:** Home cooks and chefs do not possess HPLC lab reports for amino acids in milligrams. When checked, the AI automatically computes all 9 Essential Amino Acids (Leucine, Lysine, Threonine, Histidine, Met+Cys, Tryptophan, Valine, Isoleucine, Phe+Tyr) using IFCT 2017 reference patterns calibrated to the entered protein grams and food source. If unchecked, laboratory scientists can type direct mg measurements.

Feature 3.3: The Four Interactive Diagnostic Panels

Once calculated, results are cleanly grouped into four clinical diagnostic tabs for transparent interpretation:

Panel Name	What it Evaluates & Clinical Importance
1. Protein Quality & DIAAS	Displays the weighted amino acid score (out of 10), identifies the exact primary limiting amino acid bottleneck, and shows the complementarity synergy rating.
2. Nutrient Density (PNS)	Evaluates fulfilment percentages against ICMR-NIN 2024 daily allowances for Protein, Fibre, Calcium, Iron, Zinc, Potassium, Vitamin A, Vitamin C, and B12.
3. Chronic Risk Load (NRS)	Screens recipe penalization across Sodium (Salt), Saturated Fat, Added Free Sugars, Trans Fatty Acids, Cholesterol, and High Energy Density.
4. Life-Stage Nutrition Guidance	Presents tailored clinical advisories, including anemia risk warnings, cereal-pulse synergy tips, and exact score enhancement predictions.

Feature 3.4: Dietary Optimization & Balance Tips (Clinical Advisory Engine)

This engine transforms NutriScore-India from a passive calculator into an active health optimization tool. It scans the chemical composition of the food and issues immediate, practical culinary recommendations:

- **Salt Reduction Advisory (ICMR Guideline 7):** When sodium exceeds 30% of the 2000mg upper limit (5g salt), it calculates: *'Sodium Moderation: Salt accounts for X% of daily limit. Reducing salt by 20% can improve NutriScore by +3 to +6 points.'*
- **Healthy Oil Substitution (ICMR Guideline 9):** When saturated fat is elevated, it recommends substituting vanaspati/butter with traditional cold-pressed MUFA/PUFA oils (groundnut, mustard, sesame).
- **Non-Heme Iron Bioavailability (ICMR Guideline 11):** When iron is below 20% RDA in vulnerable women or mothers, it issues a synergy prompt: *'Pairing with a Vitamin C source (lemon, amla, tomato) enhances non-heme iron absorption.'*
- **Dietary Fibre & Satiety (ICMR Guideline 3):** Flags low fiber (<2.5g) and suggests incorporating millets, sprouts, or leafy greens.

Feature 3.5: Preset Indian Recipes & Cloud History

- **1-Click Presets:** Pre-loaded with iconic Indian recipes (Moong Dal Tadka, Vegetable Khichdi, Palak Paneer, Ragi Porridge) with accurate IFCT 2017 nutrient profiles.
- **MongoDB Atlas Cloud Database:** Complete transition from SQL to MongoDB Atlas. Every analysis is automatically saved with its demographic key, grade, amino acid score, and timestamp for auditing, research, and tracking meal progression.

4. Mathematical Scoring Methodology & Calculation Steps

The PPQND scoring engine operates through an objective, multi-stage mathematical pipeline:

Step 1: Positive Nutrient Score (PNS: Scale 0 to 100)

Positive components are scored from 0 to 10 points based on demographic RDA fulfilment percentages, then weighted as follows:

$$\text{PNS} = 10 \times [(0.25 \times \text{AA_Score}) + (0.10 \times \text{Protein_Qty}) + (0.05 \times \text{Comp_Score}) + (0.10 \times \text{Fibre_Score}) + (0.15 \times \text{Minerals_Score}) + (0.15 \times \text{Vitamins_Score}) + (0.08 \times \text{Omega3_Score}) + (0.06 \times \text{PUFA_Score}) + (0.06 \times \text{MUFA_Score})]$$

Step 2: Negative Chronic Risk Score (NRS: Scale 0 to 100)

Risk factors are scored from 0 to 10 penalty points against ICMR-NIN 2024 upper chronic limits:

$$\text{NRS} = 10 \times [(0.25 \times \text{Sodium_Pts}) + (0.20 \times \text{SFA_Pts}) + (0.20 \times \text{Sugar_Pts}) + (0.15 \times \text{TransFat_Pts}) + (0.10 \times \text{EnergyDensity_Pts}) + (0.10 \times \text{Cholesterol_Pts})]$$

Step 3: Benefit-Risk Balance & Final NutriScore Equation

The overall health score reflects the biological net balance between positive density and negative disease risks. A baseline constant of 50.0 is utilized with an asymmetric 3:1 benefit-to-risk weighting ratio:

$$\begin{aligned} \text{Net Balance} &= (0.75 \times \text{PNS}) - (0.25 \times \text{NRS}) \\ \text{Final NutriScore} &= \text{Bound}(50.0 + \text{Net Balance}, \text{Min: } 0.0, \text{Max: } 100.0) \end{aligned}$$

Step 4: 7-Tier Grade Classification Matrix

Grade	Score Range	Badge Color	Clinical Classification & Health Description
A+	90.0 – 100.0	Emerald (#059669)	Optimal Therapeutic Quality: Complete amino acid spectrum, superior micronutrients, negligible chronic risk.
A	80.0 – 89.9	Green (#16a34a)	High Nutrient Density: High protein quality with complete complementarity, protective lipids, daily staple.
B	70.0 – 79.9	Lime (#84cc16)	Good Quality Staple: Balanced macronutrient and micronutrient density. Excellent foundation for Indian meals.
C	60.0 – 69.9	Amber (#eab308)	Moderate Quality: Adequate calories, but contains moderate sodium/fats or lacks optimal micronutrient balance.
D	50.0 – 59.9	Orange (#f97316)	Nutritionally Imbalanced: High refined carbohydrates, excess saturated fat, or insufficient dietary fibre.
E	40.0 – 49.9	Rose (#f43f5e)	Poor Nutritional Quality: Substantial chronic risk penalty (excess salt/sugar). Requires culinary reformulation.
F	< 40.0	Red (#dc2626)	Critical Risk Penalty: High obesogenic load, ultra-processed profile, zero protective micronutrients.

5. Staff Presentation Cheat Sheet (What to Say to Clients / Teams)

When explaining or demonstrating this system to culinary staff, dietitians, or management, highlight these 5 core messages:

1. It is 100% Indian:	Tell the team that European models like France's Nutri-Score were built for pizza, cheese, and cereal boxes. NutriScore-India is built exclusively on ICMR-NIN 2024, Indian Food Composition Tables (IFCT 2017), and Indian cooking habits.
-----------------------	---

2. It Evaluates Protein Quality, Not Just Quantity:	Emphasize that 10g of protein from dal is not utilized the same as 10g from milk or eggs. The app checks 9 Essential Amino Acids, detects the limiting amino acid, and rewards the traditional Indian wisdom of Cereal + Pulse pairing (Rice + Dal).
3. Tailored to 11 Life Stages:	Explain that a pregnant mother needs 27mg of iron, an adolescent girl needs 32mg, and an adult man needs 19mg. The system automatically recalculates health scores and warnings based on who is eating the dish.
4. It Teaches How to Cook Healthier:	Point out the 'Dietary Optimization & Balance Tips' section. It doesn't just judge a recipe; it tells the cook: 'Cut 0.5g salt to gain +4 points' or 'Squeeze lemon juice to unlock the iron in this dal'.
5. Cloud-Enabled & Ready for Scale:	Powered by MongoDB Atlas, every analyzed recipe is stored securely in the cloud, ready for recipe databases, meal planning apps, and institutional kitchen integration.

Summary: NutriScore AI provides an evidence-based, scientifically unassailable framework for modern Indian nutrition. By fusing ICMR-NIN 2024 dietary guidelines with computational food chemistry, it empowers organizations and households to make informed, healthier food choices.